

AI Scam & Manipulation Detector

Problem Statement:

With the rapid increase in online communication, scam messages have become more advanced and psychologically manipulative. Traditional systems fail to detect hidden tactics like urgency, fear, and authority pressure. This project aims to build an AI system that detects scams and explains manipulation techniques.

Flow Diagram

User Input → Preprocessing → Feature Extraction → AI Model → Psychological Detection → Output (Scam %, Explanation)

Project Folder / Repository Structure

```
AI-Scam-Detector/  
■  
■■■ data/  
■   ■■■ dataset.csv  
■  
■■■ models/  
■   ■■■ model.pkl  
■  
■■■ utils/  
■   ■■■ preprocessing.py  
■   ■■■ feature_extraction.py  
■  
■■■ app/  
■   ■■■ app.py    (Streamlit UI)  
■  
■■■ notebooks/  
■   ■■■ training.ipynb  
■  
■■■ requirements.txt  
■■■ README.md
```

Resources & Repositories

Python, scikit-learn, pandas, nltk/spaCy, Streamlit

Datasets: Kaggle SMS Spam Dataset

Tools: VS Code, GitHub

Optional: HuggingFace

Uniqueness, Limitations & Focus Areas

Uniqueness:

- Detects psychological manipulation
- Explainable AI output

Limitations:

- New scam formats
- Short messages

Focus:

- Good dataset
- Clear UI
- Fast results

Project Summary

An AI system that detects scam messages and explains manipulation techniques.
Practical, demo-friendly, and impactful for real-world use.